Map Mapping Marker

Map Mapping Marker
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1. Course Content

- After the sandbox mapping and road network annotation in the previous chapter, we have a fine digital sandbox map. But in the robot's eyes, the map itself is merely a pile of laser point clouds and obstacle boundaries — it does not understand where the "shipping area" is, where "Shelf No. 1" is, or where the "kitchen waste bin" is.

The core mission of **Map Mapping (Semantic Map Mapping)** is to build a bridge between the human natural language world and the robot's mathematical space coordinates, linking semantic places with spatial poses:

- **Human perspective**: Natural language instructions only need to say "go to the shipping area" or "head to Shelf No. 3";
- **Robot perspective**: After receiving the instruction, quickly look up the corresponding global world coordinates in the `map_mapping.yaml` mapping dictionary

  $$(X, Y, Z)$$

  and the quaternion heading;

  $$(Q_x, Q_y, Q_z, Q_w)$$

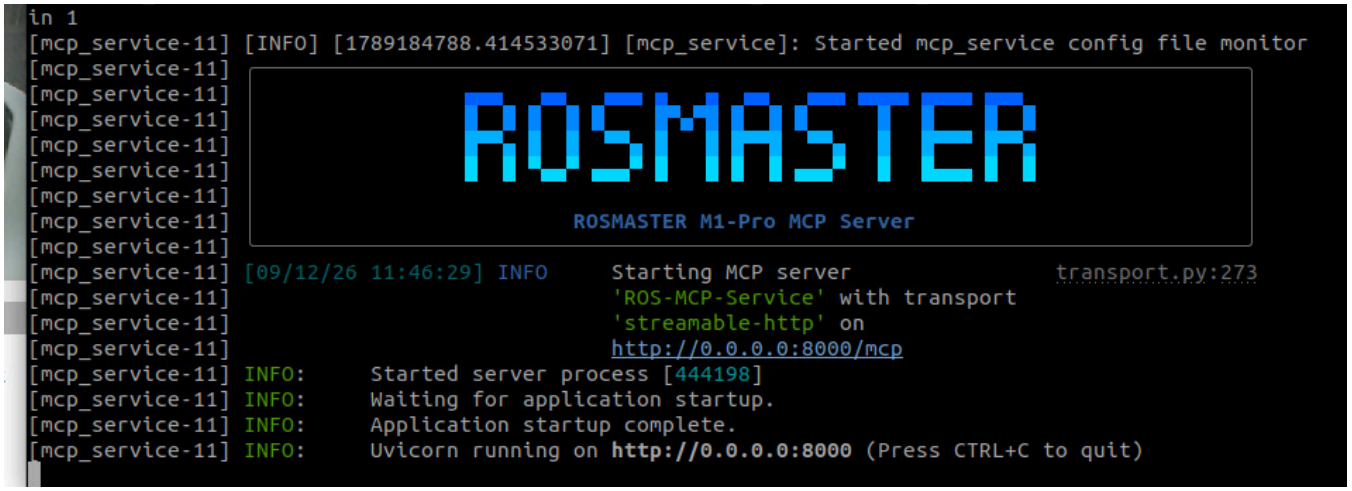
> **Important**
>
> - This lesson is a comprehensive lesson, demonstrating and explaining the use and operation steps of the sandbox map. For detailed parameter debugging and source code explanation, refer to the preceding lesson: [OpenClaw Slam Mapping and Navigation].

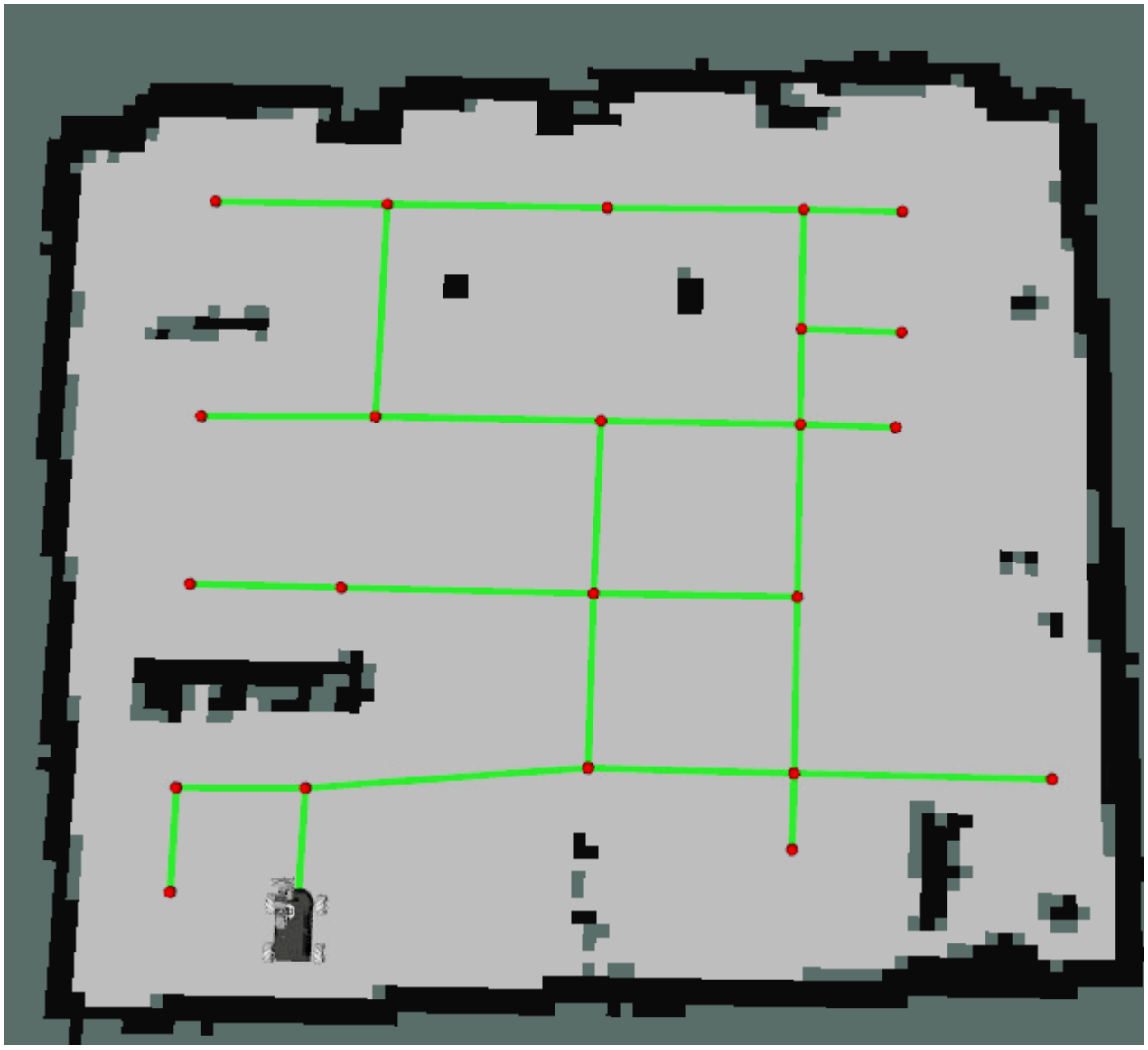
2. Preparation

2.1 Map Preparation

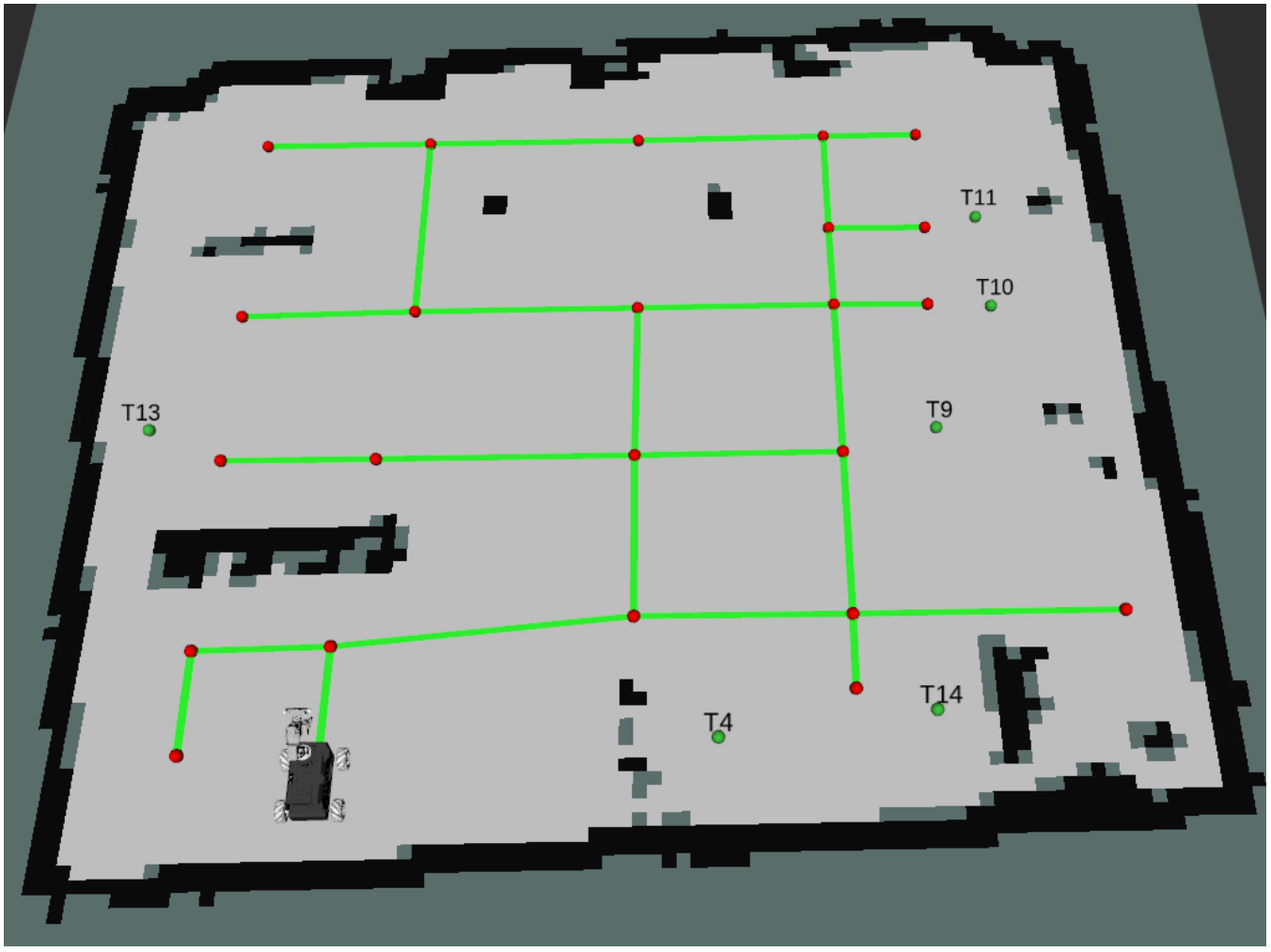
- First follow the [Building the Sandbox Map] lesson to complete building the sandbox map, pose map, and ArUco visual relocalization markers (optional). If the relocalization function is not needed, you can cancel the startup parameter `relocalization:=true`.

```
ros2 launch yahboomcar_bringup mcp_service.launch.py use_camera:=true use_nav:=true
relocalization:=true
```





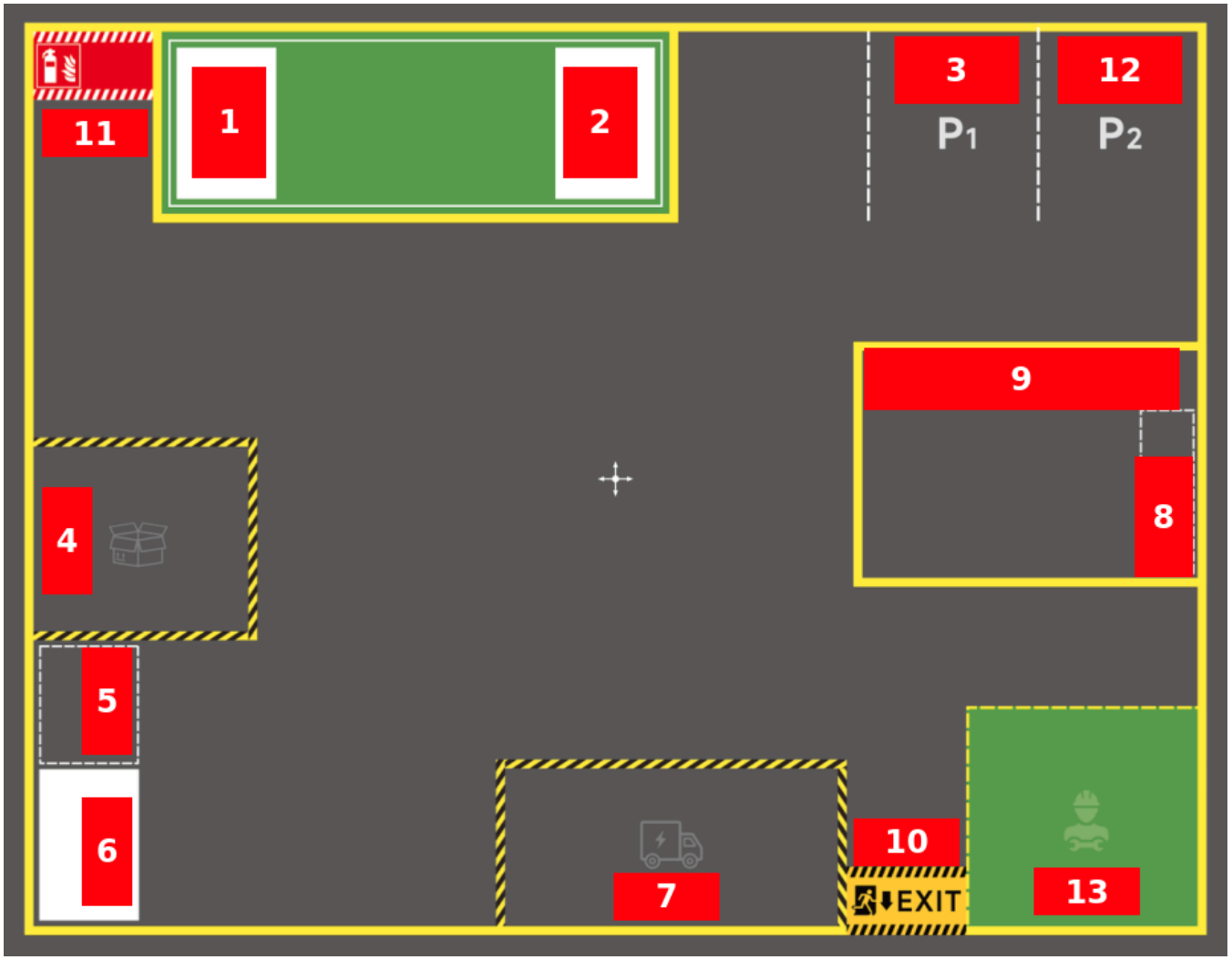
- After the visual relocalization function is started, the markers recorded in the [Visual Relocalization Marker] chapter can be seen in the map after the program has run for a while, as shown in the figure below:



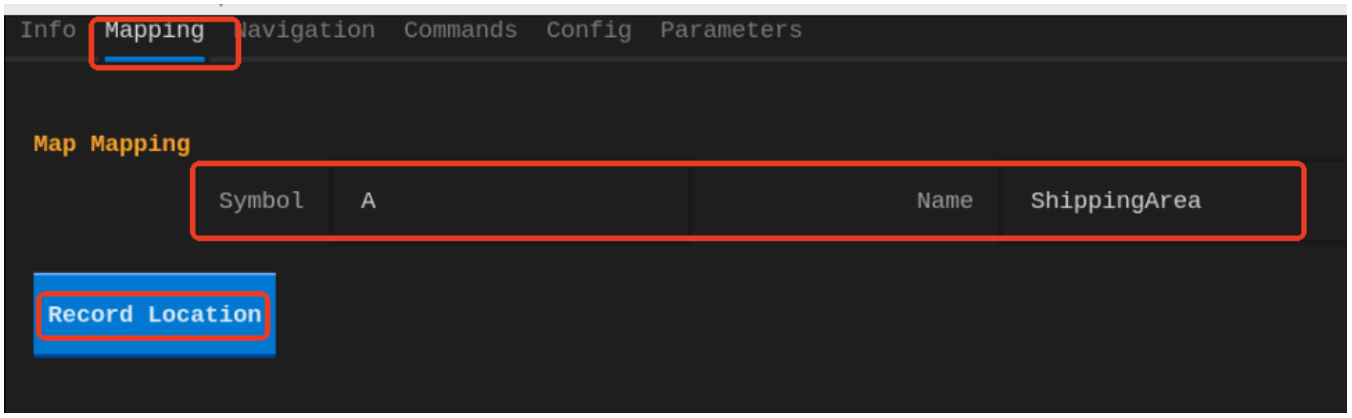
3. Recording the Map Mapping

- The placement of the map site and the area naming are arbitrary and can be modified according to needs. The following is a reference area naming case:

1. Shelf No. 1
2. Shelf No. 2
3. Parking lot No. 1
4. Goods warehouse
5. Goods sorting area
6. Shelf No. 3
7. Shipping area
8. Waste sorting area
9. Trash can positions: kitchen waste bin, recyclable bin, other waste bin, hazardous waste bin
10. Safety exit
11. Fire escape



- Refer to the semantic place recording operation process in [OpenClaw-slam Mapping and Navigation], and record all semantic places into the map mapping file through the RobotManager interface or CLI commands.

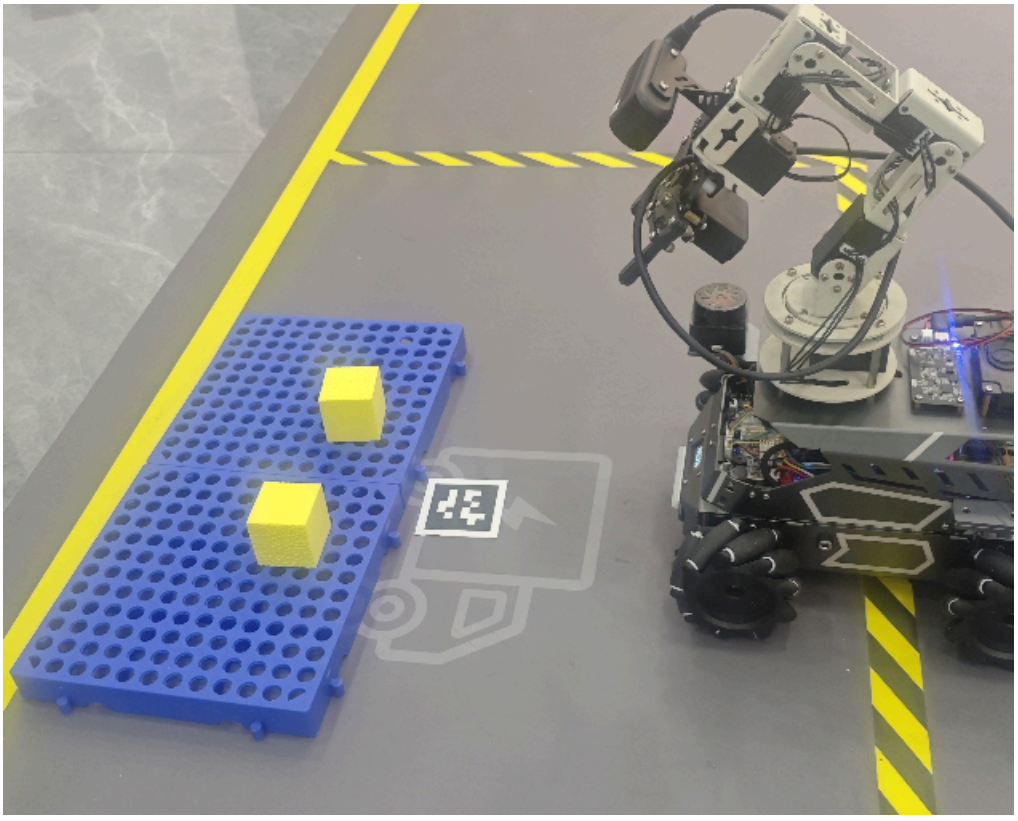


Operation Process Notes

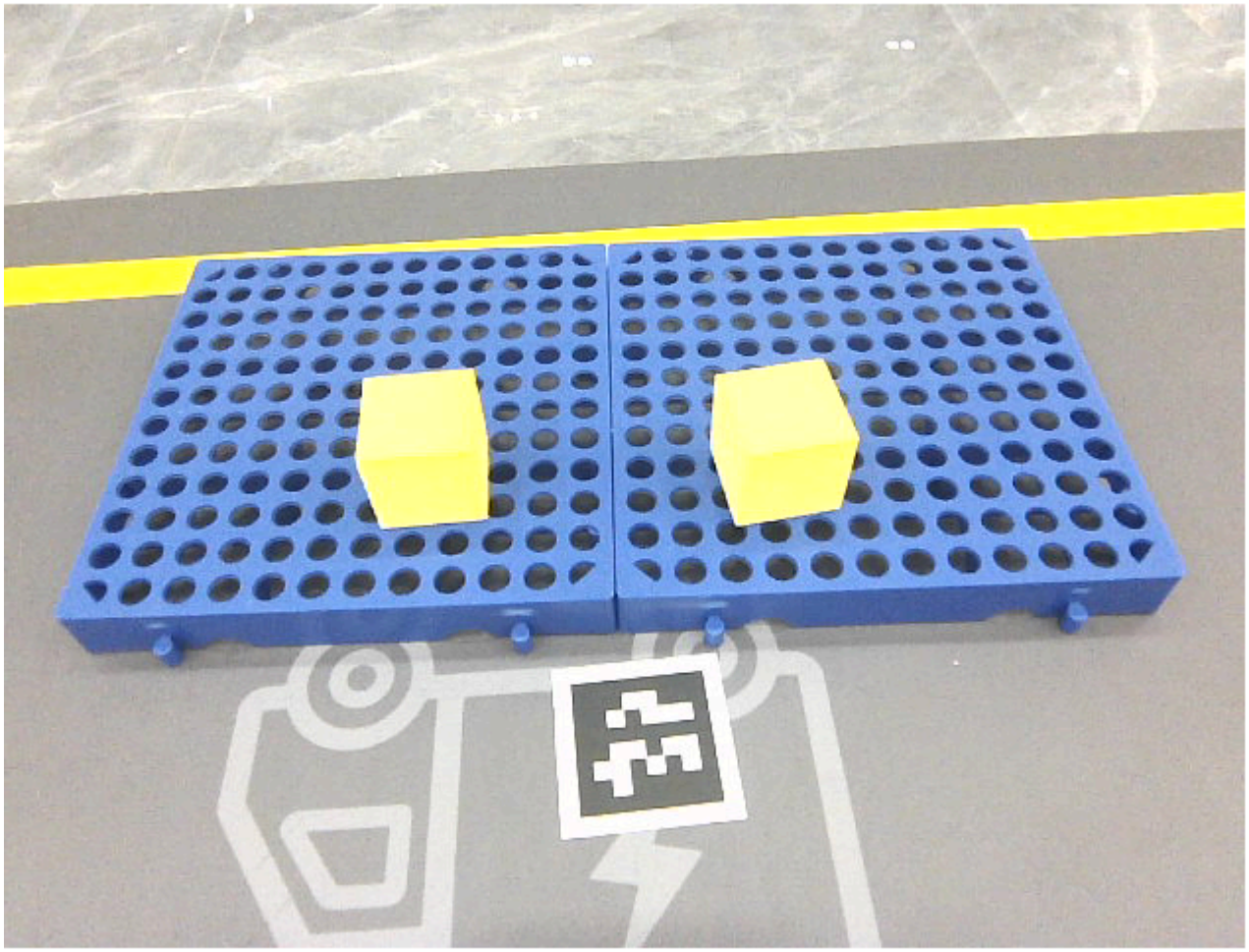
- **Important**: Before recording a position, be sure to check whether the laser contour matches the environment edge features. If there is a deviation, first use the **2D Pose Estimate** tool to perform manual relocalization before recording the position.

Recording example:

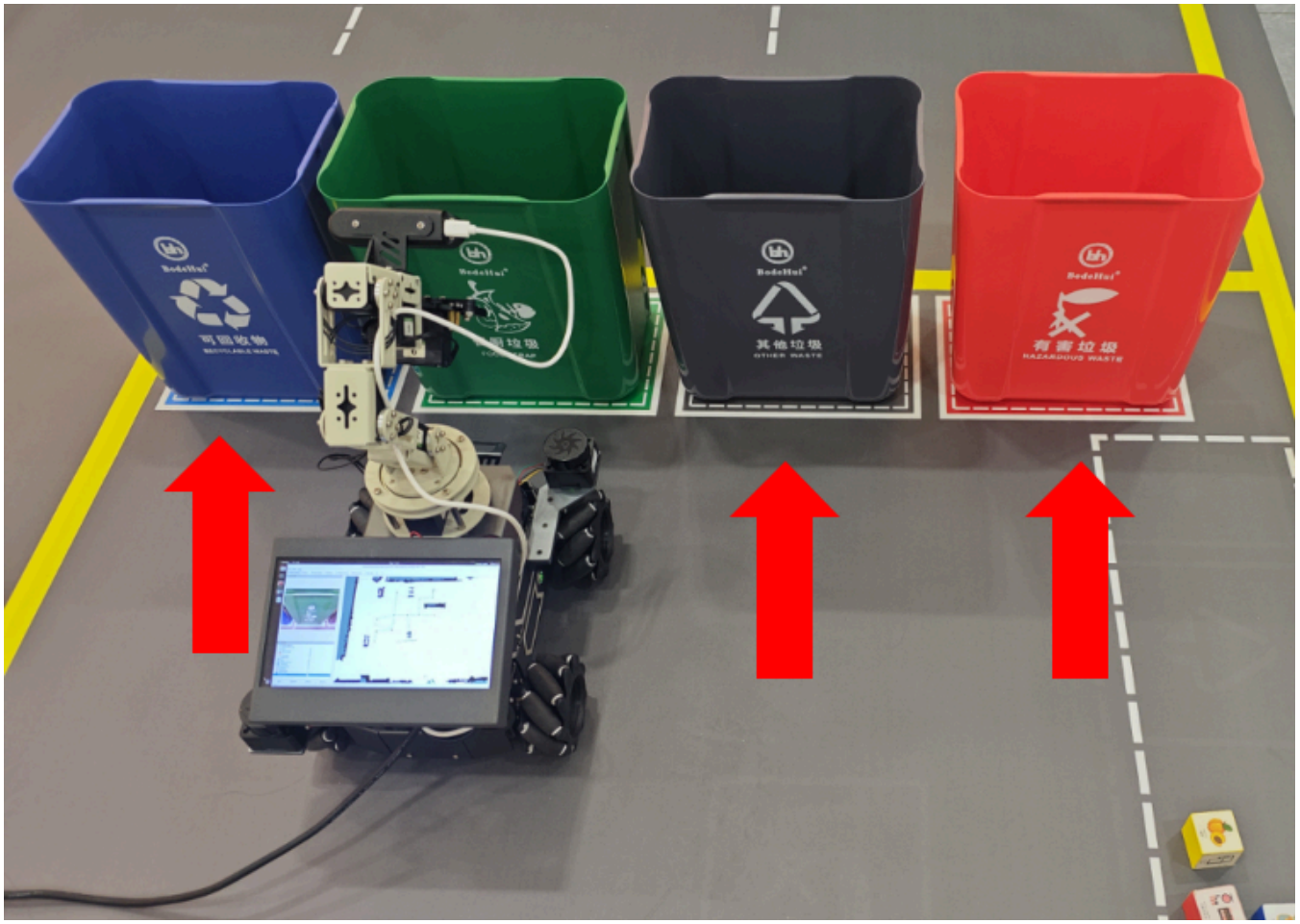
- Record the shipping area: move the car to the place near the shipping area where the position needs to be recorded.

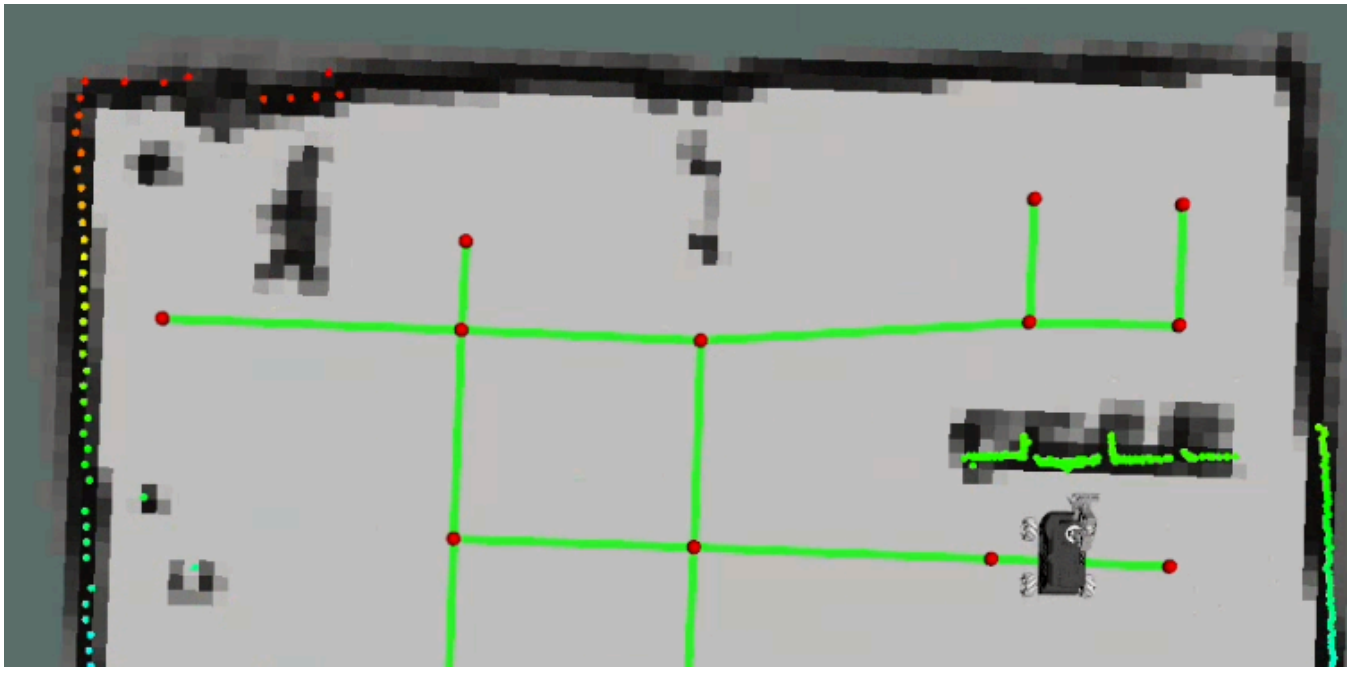


- Ensure the goods in the area can be fully seen in the field of view.

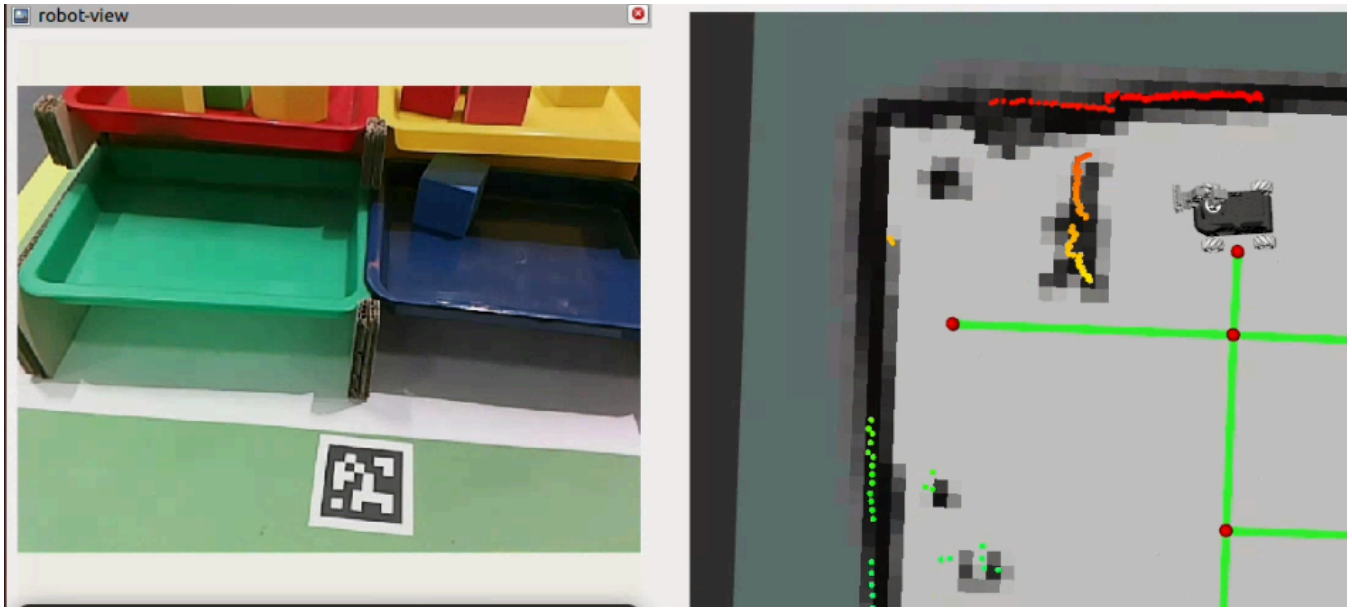


- **Record the trash can positions**
  - The positions of different types of trash cans need to be recorded separately. Place the car in front of the trash can to be recorded, then record.
  - For example, to record the kitchen waste bin:
    - The corresponding trash can is later looked up according to the waste category:
    - HazardousWaste → hazardous waste
    - FoodWaste → kitchen waste
    - OtherWaste → other waste
    - RecyclableWaste → recyclable waste

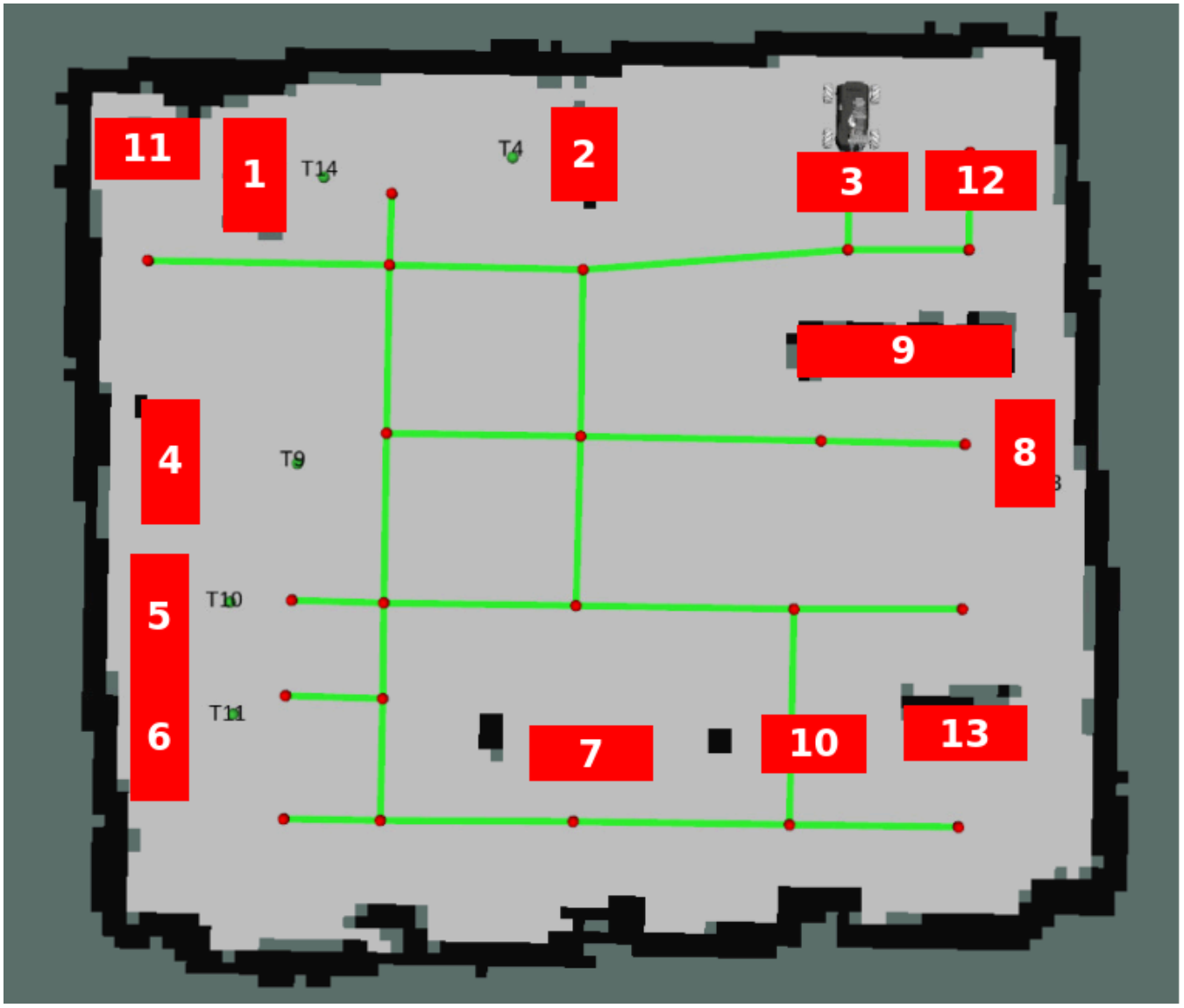




- When recording the shelf position, ensure the tag marker and the shelf can be seen in the field of view.



- The following is a schematic diagram of all semantic places in the grid map.



- View the map mapping file `map_mapping.yaml`. The file path is:

```
~/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_bringup/param/map_mapping.yaml
```

- Check whether all the positions that need to be used have been recorded.
- At this point, the entire sandbox map has been thoroughly endowed with embodied spatial semantics that are "human-readable and machine-interpretable". The robot is now fully ready to accept complex factory comprehensive task scheduling!

